DAY - 12 SEAT NUMBER | M | 0 | C | 3 | 4 | 8 |

2012	III	14	1100	V – 634	(E)

COMPUTER SCIENCE
PAPER - 1 (D-9)

Time : 3 Hours	4 Pages	Max. Marks : 50

Instructions : (1) *All questions are compulsory.*

(2) *Figures to the right indicate full marks.*

(3) *Use of any type of calculator is not allowed.*

(4) *Header files and comments in C++ program are not necessary.*

(5) *Draw neat diagram wherever necessary.*

1. (A) Select correct alternatives and rewrite the following : 1

(a) _____ tag is used to put a horizontal rule in HTML Code.

(i) <HR>

(ii)

(iii) <P>

(iv) <TD>

(b) Maximum Number of Nodes of Symmetric Binary Tree with depth n are _____ 1

(i) 2^n

(ii) $2^n - 1$

(iii) $\log_2 n$

(iv) n^2

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(c) Linux is a _____ Software. 1

(i) Public Domain

(ii) Free

(iii) Paid

(iv) Private

(d) Out of the following C++ operators, _____ operator can be overloaded. 1

(i) Sizeof

(ii) ::

(iii) ?:

(iv) *

(B) Answer **any two** of the following :

(a) What is GUI ? State any four advantages of GUI. 3

(b) Explain use of the following HTML Tags with example : 3

(i) <Pre>

(ii)

(iii) <Sub>

(c) Explain the following data structures with suitable diagrams : 3

(i) Linear Array

(ii) Linked List

(iii) Tree

2. (A) Answer **any two** of the following :

(a) Explain Insertion and Extraction Operators in C++. 3

(b) Explain in brief any six Data Structure Operations. 3

(c) What are Pointer Arrays ? Explain giving an example. 3

(B) Answer **any one** of the following :

(a) What are the different functions performed by Memory Management in an Operating System ? Explain types of Partitioning in brief.

(b) Describe with an example, how member functions of a class can be defined Outside the Class Definition and Inside the Class Definition ?

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(A) Answer **any two** of the following :

(a) What is the difference between a Worm and a Virus ? Explain how these can be prevented ? 3

(b) State any six principal advantages of Object Oriented Programming. 3

(c) Explain difference between Operator Function as a Member Function and as a Friend Function. 3

(B) Answer **any one** of the following :

(a) Define a block with reference to Operating Ssystem. What are the parameters of a block of data that are concerned with an Operating System. Explain in brief. 4

(b) Write Bubble Sort Algorithm with a suitable example. 4

(A) Answer **any two** of the following :

(a) Explain with flow charts the following control structures : 3

(i) Sequence Logic

(ii) Selection Logic

(iii) Iteration Logic

(b) With a suitable example show representation of linked list in Memory. 3

(c) What is a Constructor and a Destructor ? Give one example for each. 3

(B) Answer **any one** of the following :

(a) What are the classes in C++ for File Stream Operation ? How do you Open and Close File in C++ ? Mention any four File Modes. 4

(b) What is a Process ? Explain ready, running and blocked process states in Process Management. 4

Answer **any two** of the following :

(a) Write a C++ program that will read a line of text and count number of words in a text. 5

(b) Write a program in C++ to find GCD and LCM of two inputted numbers using methods enter( ) and compute( ). 5

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(c) Write exact output of the following HTML Code with font specifications in brackets : 5

```
<HTML>
<MEAD>
<TITLE> Computer Step
</TITLE>
</HEAD>
<BODY>
<H2> <U> Title : Megastar Company Ltd. </U> </H2>
<P> Address : Surya Complex, Delhi </P>
<H4> <B> Deals In : </B> </H4>
<UL>
<LI> Software
<LI> Hardware
</LI> Peripherals
</UL>
</BODY>
</HTML>
```

OR

5. Answer **any two** of the following :

(a) Write a C++ program to accept a number and test whether it is prime or not. 5

(b) Implement a point class for two dimensional points (x, y). Include a function dist( ) to return points distance from origin (0, 0) and a display( ) function to print result. 5

(c) Write MTML Code for the following : 5

Roll No.	Name	Marks		Total
		CS1	CS2	
1	RINA	75	70	145
2	RONAK	80	90	170
3	TINA	70	85	155

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